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REVIEW

Effect of primary ovarian insufficiency and early natural menopause on mortality: a meta-analysis

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ABSTRACT

Objective The aim of this review was to systematically evaluate the associations of all-cause, cardiovascular and all-cancer mortality with primary ovarian insufficiency (POI) and early natural menopause (ENM).

Methods Electronic databases for relevant studies were searched up to February 28, 2015. POI and ENM were usually defined as spontaneous menopause before age 40 years and at age 40–44 years, respectively.

Results A total of nine articles were derived from seven prospective cohort studies. In all studies, age of menopause was self-reported. Our meta-analysis showed that POI women had a higher risk of death from all causes (pooled relative risk (RR) 1.39, 95% confidence interval (CI) 1.10–1.77) and ischemic heart disease (IHD) (pooled RR 1.48, 95% CI 1.02–2.16) when compared with women at normal age at natural menopause (ANM). No significant association was detected from stroke and all-cancer mortality between POI women and normal ANM women. Only a slightly higher risk of death from IHD (pooled RR 1.09, 95% CI 1.00–1.18) was found when ENM women were compared with normal ANM women.

Conclusion The results of our study demonstrated that POI was associated with a higher risk of IHD and all-cause mortality; ENM was only associated with a slightly higher risk of IHD mortality.

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Primary ovarian insufficiency; early natural menopause; cardiovascular disease; cancer; mortality; meta-analysis

Introduction

Natural menopause is defined as the permanent cessation of menstruation due to natural loss of ovarian follicular function[1]. The median age at natural menopause (ANM) has mostly been estimated at between 50 and 53 years[2–4]. Spontaneous menopause before age 40 is defined as primary ovarian insufficiency (POI)[5] and occurs in 0.9–1.2% of females[6]. It depends upon the age at diagnosis; the probability of a genetic, idiopathic or autoimmune cause is more or less likely[7,8]. Spontaneous menopause at age 40–44 years is defined as early natural menopause (ENM)[9] and occurs in approximately 5% of females.

ANM may be a marker not only of reproductive aging but also of general health and somatic aging[10]. Accumulating evidence suggests that early menopause is associated with premature death, including death from cardiovascular disease (CVD) and cancer, but the results are not entirely consistent and the magnitudes of the relationships also varied among studies[11–14]. Compared with women who experience early surgical menopause, ENM women may experience a different effect on health and mortality[15,16]. The further classification of early menopause should be considered. Recently, a meta-analysis summarized data from prospective cohort studies and reported that POI could increase the risk of CVD. However, that meta-analysis did not examine the outcomes of cancer and all-cause mortality. Our objective was to systematically review and quantitatively estimate the relationship between POI, ENM and all-cause, cardiovascular and all-cancer mortality based on a meta-analysis.

Methods

Search strategy and study selection

Electronic databases (EMBASE, MEDLINE and the Cochrane Library) were searched to February 28, 2015 using the following terms: endogenous estrogen exposure or menopause and mortality or death or fatal. The search was limited to humans and was restricted to all English-language published studies. Studies were included if they fulfilled the following criteria: (1) accrued more than 1 year of follow-up; (2) assessed age at natural menopause of the general population at baseline; (3) recorded all-cause mortality and/or

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cardiovascular mortality and/or all-cancer mortality; and (4) reported the multivariate-adjusted relative risks (RRs) and 95% confidence intervals (CIs) for mortality associated with POI (menopause at <40 years) versus reference (menopause at ≥45 years), or reported RRs and 95% CIs for ENM (menopause at 40–44 years) versus reference (menopause at ≥45 years), respectively. However, the cut-off points for POI and ENM were not completely consistent; ≤40 years for POI and 41–43 years or 41–44 years for ENM were also accepted. We excluded studies that assessed non-fatal cardiovascular events and studies on subjects with unnatural menopause. Except for ischemic heart disease (IHD) and stroke mortality, studies reporting only specific mortality (e.g. cervical cancer mortality) were also excluded.

Quality evaluation

Quality assessment was based on the Newcastle-Ottawa Scale for cohort studies[17]. The scale consists of eight items, of which seven were applicable to our study. Items are grouped into three domains including selection, comparability, and outcome. Two independent reviewers read and scored each of the included studies. We assigned total scores of 0–3, 4–5, and 6–8 stars for low-, moderate-, and high-quality studies, respectively[18]. Any discrepancies between the reviewers were resolved with a joint reassessment until a consensus was reached.

Data extraction

A predetermined set of data was extracted from each study that included the name of the first author, year of publication, study name, country of study, sample size, age range, length of follow-up, RRs with corresponding 95% CIs for all categories of age at natural menopause, outcomes (i.e. CVD, IHD, stroke, all-cancer and all-cause mortality), adjusted factors of smoking and hormone therapy, and other adjusted factors. The outcomes of CVD mortality in some studies included both IHD and stroke mortality. We used the results of the original studies from multivariable models with the most complete adjustment for potential confounders. Of the studies, one study (The Adventist Health Study) published two articles, but reported different outcome variables (one reported all-cause and IHD mortality and the other reported all-cause, IHD, and stroke mortality) and different follow-up periods; we extracted data of longer follow-up or the most informative one. In one study[14], never-hormone therapy users and never-smokers were analyzed separately; we extracted data of the former one due to the hormone therapy being a stronger confounder than smoking in the results.

Data analysis

Multivariate-adjusted outcome data (expressed as RRs and 95% CIs) were used to estimate the summary statistics. These values were logarithmically transformed in each study and the corresponding standard errors (SEs) were calculated. The statistical analysis used the inverse variance approach to combine log RRs and SEs. A random-effects model was used as this can provide more conservative results than a fixed-effects model[19]. A separate analysis using a fixed-effects model was also conducted and, unless otherwise stated, no differences in the summary estimates were found.

We used the I^2 statistic to assess heterogeneity. Low, moderate, and high degrees corresponded to I^2 values of 25%, 50%, and 75%, respectively[20]. The influence of an individual study on the summary estimates was investigated by sensitivity analysis in which the meta-analysis estimates were computed omitting one study in turn. Moreover, we investigated potential sources of heterogeneity by conducting subgroup analyses. Results are presented as pooled RR and their 95% CIs. We generated Forest plots sorted by level of precision to visually assess relative risks across studies. p Values were two-tailed, and the statistical significance was set at 0.05. Finally, Egger's test was used to assess publication bias[21]. All analyses were performed with Stata, version 10.0, software (Stata Corp LP, College Station, Texas, USA).

Results

Selected studies and characteristics

Through electronic searches, we identified 3287 unique articles. Of these, 3244 were excluded on the basis of titles and abstracts, leaving 43 articles for further evaluation. Finally, nine articles were included in the review after screening full texts (Figure 1). Primary analysis of the nine articles included was derived from seven prospective cohort studies[10,11,14,22–27]; six reported all-cause mortality; five reported all CVD or both IHD and stroke mortality; four reported IHD mortality; five reported stroke mortality; and four reported cancer mortality, respectively.

Of the seven cohort studies, four were from the United States, two from Asia and one from Europe. The nine articles were published between 1989 and 2013. The mean follow-up duration in the studies ranged from 4 to 20 years, and the sample sizes ranged from 3191 to 267 400 women. Ascertainment of mortality outcomes was based on record linkage with official death certificates, and International Classification of Diseases

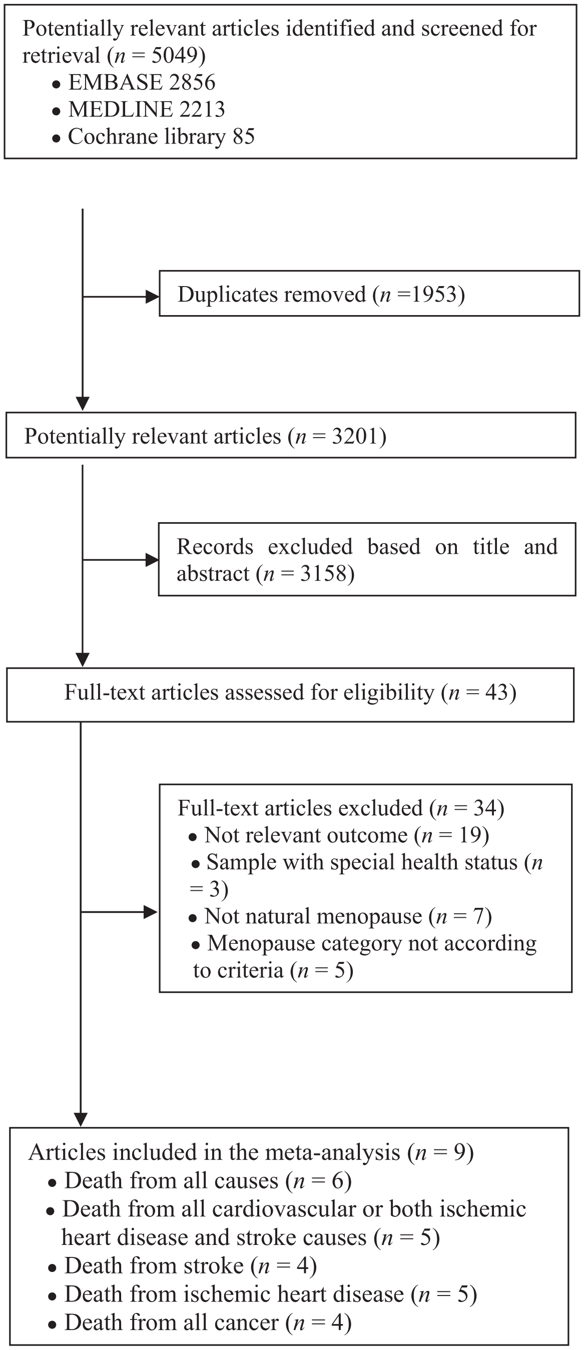

Figure 1. Flow diagram of the search results for the meta-analysis.

codes were used to identify CVD mortality. In all studies, age of menopause was self-reported. The reference group of age at menopause was in the range of 49–54 years old, with the exception of two studies[23,27]. The factors of smoking and hormone therapy were controlled for or excluded in most studies. All of the included studies were of adequate quality. Two studies received five stars, indicating moderate quality, and five studies received six to eight stars, indicating high quality. The details and characteristics of included studies are listed in Table 1.

POI, ENM and all-cause mortality

Of the six studies, four were from the United States, one was from Norway, and one was from Japan. Meta-analysis showed that POI women had a higher risk of death from all causes when compared with normal ANM women (pooled RR 1.39, 95% CI 1.10–1.77) for five studies. The I^2 was 78.7%, suggesting a high degree of heterogeneity between studies (Q statistic, $p = 0.001$) (Figure 2). Sensitivity analyses showed that none of the individual studies dramatically influenced the pooled RR (RR range: 1.27 (95% CI 1.02–1.59) to 1.58 (95% CI 1.20–2.08)). Egger's test was significant ($p = 0.017$), indicating that publication bias existed. Subgroup analyses results indicated that non-significant RRs were found from studies conducted by the Norwegian group, and no control was made for groups with smoking factor and hormone therapy factor (Table 2).

There was no significant association for risk of death from all causes between normal ANM women and ENM women. Subgroup analyses indicated no significant association in all the groups with the exception of a very slight association found in one study where the smoking population was excluded (Table 2), and no publication bias was detected ($p = 0.843$).

POI, ENM and risk of CVD mortality

Total CVD mortality

Five studies reported total CVD mortality or both IHD and stroke mortality. Four studies were from the United States, and one was from China. By meta-analysis, the RR for total CVD deaths comparing the POI women versus normal ANM women was 1.24 (95% CI 0.98–1.58; $I^2 = 0\%$). The association for risk of death from CVD between ENM and normal ANM women was not significant (Figure 3). None of the subgroup analyses results were significant for POI and ENM (data not shown). No significant evidence of publication bias was observed ($p = 0.530$ for POI, $p = 0.880$ for ENM).

IHD mortality

Three articles for POI women and four articles for ENM women reported death from IHD. Meta-analysis showed that POI women had a higher risk of death from IHD (pooled RR 1.48, 95% CI 1.02–2.16) when compared with normal ANM women, and ENM women had a slightly higher risk of death from IHD (pooled RR 1.09, 95% CI 1.00–1.18) when compared with normal ANM women (Figure 4). After excluding one study for very high weight (94.18%), the pooled RR became non-significant

Table 1. Characteristics of studies included in the meta-analysis.

Study	Study name	Location	Mean age (years) (range or SD)	Follow-up (years)	n	Endpoints	Age at menopause (years)	Controlled factors: Smoking	Controlled factors: Hormone therapy	Controlled factors: Other factors	NOS scale
Li, 2013[14]	BWHS	USA	38 (21–69)	13	11 052	all-cause, cancer, CVD mortality	<40, 40–44, 45–49, 50–54 (ref), ≥ 55	yes	exclude	age, education, marital status, BMI, alcohol consumption, physical activity, dietary pattern, age at menarche, parity, age at first birth, oral contraceptive use, lactation duration, and unilateral oophorectomy	6
Gallagher, 2011[22]	Textile Workers Study	China	>30	9–11	267 400	IHD, stroke mortality	<40, 40–44, 45–49, 50–54 (ref), ≥ 55	exclude	no	age	5
Amagai, 2006[23]	Jichi Medical School Cohort Study	Japan	61 (36–89)	9.2 (average)	3824	all-cause mortality	<40, 40–44, 45–49 (ref), 50–54, ≥ 55	yes	no	age, systolic blood pressure, total cholesterol, high density lipoprotein cholesterol, history of diabetes mellitus, BMI, alcohol drinking habits, marital status, study area	6
Mondul, 2005[11]	Cancer Prevention Study II	USA	≥ 30	20	68 154	all-cause, cancer, IHD, stroke mortality	40–44, 45–49, 50–54 (ref)	exclude	exclude	age, race, marital status, BMI, age at menarche, parity, education, alcohol consumption, oral contraceptive use, and exercise	8
Jacobsen, 2004[24]	Norwegian Women Cohort Study	Norway	48.4 (32–74)	20.6 (average)	19 731	stroke mortality	≤ 40, 41–43, 44–46, 47–49, 50–52 (ref), 53–55, 56–60	no	no	age, county, occupational group, and birth cohort	7
Jacobsen, 2003[25]	Norwegian Women Cohort Study	Norway	48.4 (32–74)	20.6 (average)	19 731	all-cause mortality	≤ 40, 41–43, 44–46, 47–49, 50–52 (ref), 53–55, ≥ 56	no	no	age, county, occupational group, and birth cohort	7
Jacobsen, 1999[26]	Adventist Health Study	USA	≥ 25	12	6182	all-cause mortality	35–40, 41–44, 45–48, 49–51 (ref), 52–55, 56–60	yes	yes	coronary heart disease, diabetes or hypertension at start of follow-up, BMI and physical activity	5
						IHD mortality	35–40, 41–44, 45–48, 49–51 (ref), 52–55, 56–60	no	exclude	diabetes, hypertension, parity, age at first birth, and physical activity in leisure	
Cooper, 1998[27]	NHANES	USA	50–86	4	3191	all-cause, cancer, IHD, and cerebrovascular disease mortality	<40, 40–44, 45–49, ≥ 50 (ref)	yes	yes	age at baseline, years of follow-up, race, education	7
Snowdon, 1989[10]	Adventist Health Study	USA	≥ 25	6	3334	cancer, IHD, and stroke mortality	<40, 40–44, 45–49, 50–54 (ref), ≥ 55	yes	yes	age, Quetelets index, education, marital status, age at first live birth, live births, miscarriages and stillbirths	6

ref, reference; CVD, cardiovascular disease; IHD, ischemic heart disease; BMI, body mass index; NOS, Newcastle-Ottawa Scale; SD, standard deviation.

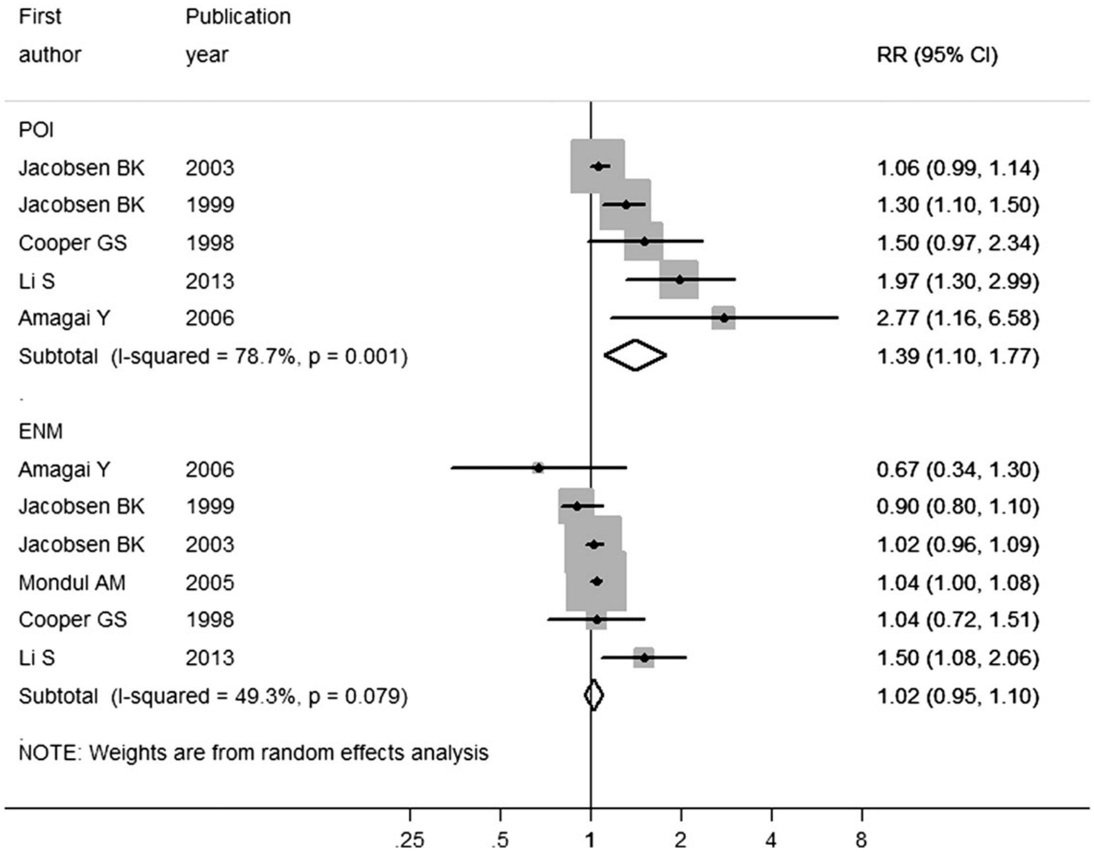

Figure 2. Forest plot of all-cause mortality comparison: early age at natural menopause vs. normal age at natural menopause. POI, primary ovarian insufficiency; ENM, early natural menopause; RR, relative risk.

Table 2. Subgroup analyses for all-cause mortality ($n = 6$).

Source	Number of reports	Relative risk (95% confidence interval)	p Value	Heterogeneity p Value	Heterogeneity I^2 (%)
Primary ovarian insufficiency					
Location					
USA	3	1.475 (1.157–1.881)	0.002	0.173	43.1
Japan	1	2.770 (1.163–6.597)	0.021	–	–
Norway	1	1.060 (0.988–1.137)	0.105	–	–
Reference					
50–54 years	3	1.293 (1.009–1.657)	0.042	0.002	84.5
Other	2	1.810 (1.040–3.151)	0.036	0.217	34.5
Smoking					
Controlled	4	1.580 (1.201–2.079)	0.001	0.115	49.4
Not controlled	1	1.060 (0.988–1.137)	0.105	–	–
Hormone therapy					
Controlled	2	1.321 (1.141–1.529)	0.001	0.548	0.0
Not controlled	2	1.548 (0.617–3.886)	0.352	0.031	78.6
Excluded	1	1.970 (1.299–2.988)	<0.001	–	–
Early natural menopause					
Location					
USA	4	1.050 (0.906–1.217)	0.223	0.044	62.9
Japan	1	0.670 (0.343–1.310)	0.242	–	–
Norway	1	1.020 (0.957–1.087)	0.541	–	–
Reference					
50–54 years	4	0.916 (0.619–1.354)	0.279	0.018	75.3
Other	2	1.049 (0.962–1.144)	0.659	0.470	0.0
Smoking					
Controlled	4	1.030 (0.771–1.376)	0.461	0.028	67.0
Not controlled	1	1.020 (0.957–1.087)	0.541	–	–
Excluded	1	1.040 (1.001–1.081)	0.046	–	–
Hormone therapy					
Controlled	2	0.921 (0.795–1.066)	0.599	0.482	0.0
Not controlled	2	0.949 (0.696–1.295)	0.741	0.221	33.1
Excluded	2	1.204 (0.847–1.712)	0.300	0.027	79.5

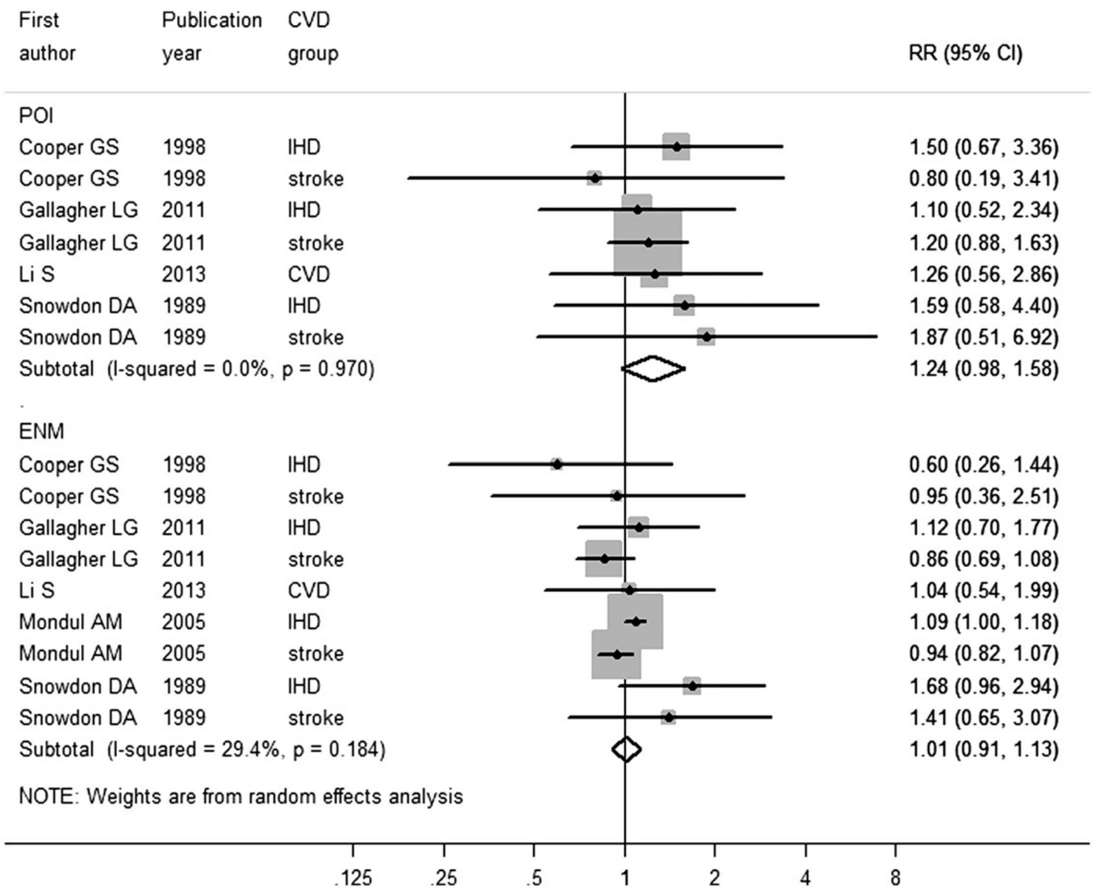

Figure 3. Forest plot of cardiovascular mortality comparison: early age at natural menopause vs. normal age at natural menopause. POI, primary ovarian insufficiency; ENM, early natural menopause; CVD, cardiovascular disease; IHD, ischemic heart disease; RR, relative risk.

(pooled RR 1.01, 95% CI 0.73–1.41). No publication bias was detected ($p = 0.490$ for POI, $p = 0.457$ for ENM).

Stroke mortality

Four articles for POI women and five articles for ENM women reported death from stroke. Meta-analysis of the estimates for risk of death from stroke showed no significant effect for POI women or ENM women when compared with normal ANM women (Figure 5). No significant evidence of publication bias was observed ($p = 0.495$ for POI, $p = 0.545$ for ENM).

POI, ENM and risk of all-cancer mortality

A total of four studies reported death from all cancers. All studies were from the United States, and had controlled for the confounders of smoking and hormone therapy or excluded them at baseline. Meta-analysis of the estimates for risk of death from all cancer showed no significant effect for POI or ENM women versus normal ANM women (Figure 6). A high degree of heterogeneity ($I^2 = 77.0\%$, $p = 0.005$) was observed in ENM women versus normal ANM women. But the heterogeneity was not significantly reduced after exclusion of any single study. The test for publication bias was not significant ($p = 0.164$ for POI, $p = 0.245$ for ENM).

Discussion

The present study demonstrated that POI was associated with a 48% higher risk of IHD mortality and a 39% higher risk of all-cause mortality when compared with normal ANM, and perhaps had a 24% higher risk of CVD mortality. No difference was found from stroke and all-cancer mortality in POI women versus normal ANM women. Only a slightly higher risk of death from IHD was detected when comparing ENM women with ANM women.

The association of POI with risk of death from all causes is biologically plausible, but the potential mechanisms remain unclear. Some biological mechanisms would support the observed association between POI and risk of IHD and all-cause mortality. Estrogen receptors are found in most tissues including vascular endothelial cells, smooth muscle cells, intestinal

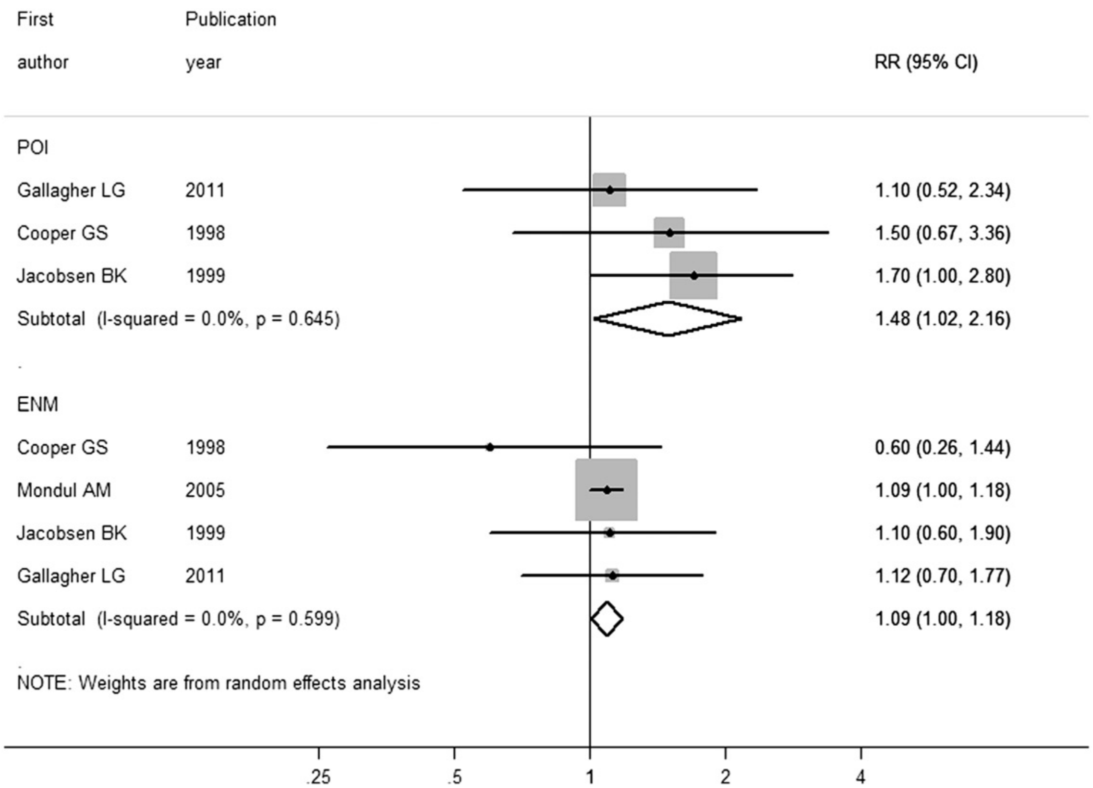

Figure 4. Forest plot of ischemic heart disease mortality comparison: early age at natural menopause vs. normal age at natural menopause. POI, primary ovarian insufficiency; ENM, early natural menopause; RR, relative risk.

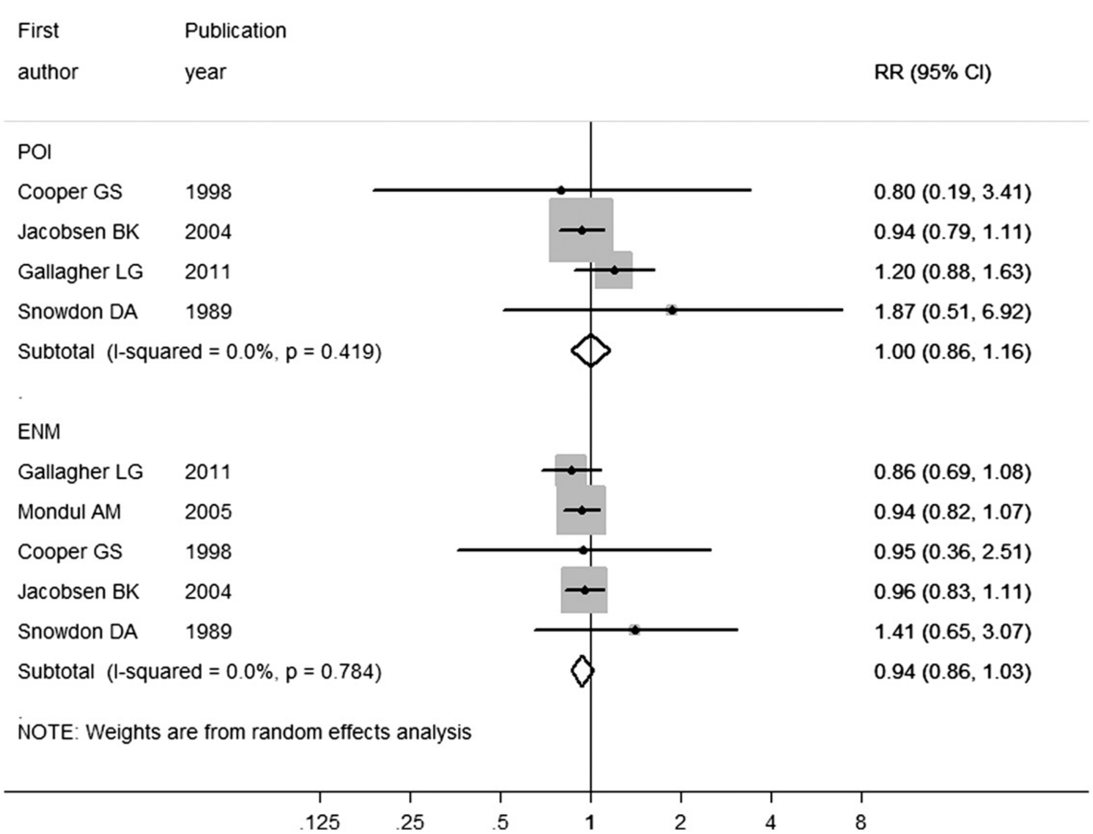

Figure 5. Forest plot of stroke mortality comparison: early age at natural menopause vs. normal age at natural menopause. POI, primary ovarian insufficiency; ENM, early natural menopause; RR, relative risk.

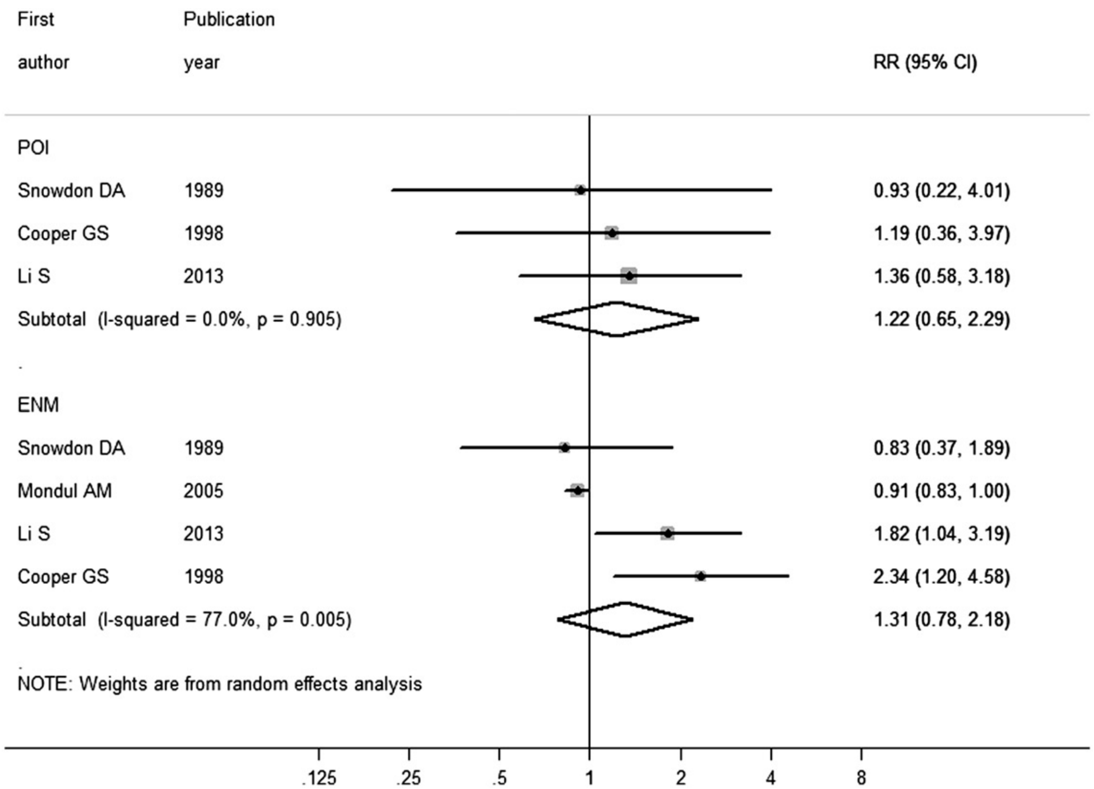

Figure 6. Forest plot of all-cancer mortality comparison: early age at natural menopause vs. normal age at natural menopause. POI, primary ovarian insufficiency; ENM, early natural menopause; RR, relative risk.

mucosa, endometrium, breast cancer cells, ovarian stroma and brain[28,29], indicating that all of these tissues may be affected by a decrease in estrogen in postmenopausal women. Estrogen may also play a role in the maintenance of immune function; it has been shown that women who undergo surgical menopause develop impaired immune function[30]. Calculated data from the randomized, controlled trial by Rossouw and colleagues have clearly demonstrated that postmenopausal hormone therapy reduces both IHD and total mortality in women who were within 10 years of menopause and under the age of 60 years[31].

In our analysis, although POI was associated with a higher risk of IHD mortality, the association was not significant for stroke mortality. This result is consistent with a report from the Nurses' Health Study which focused on ANM and risk of CVD[32] and a review of CVD and POI[33]. Early menopause as a risk factor for IHD has been the subject of many previous studies, but the relationship between early menopause and stroke is still inconsistent. Studies indicate that early menopause increases the risk of stroke, in particular ischemic stroke[34]. Since the pathogenetic mechanisms of ischemic and hemorrhagic stroke are different, future researchers should consider ischemic stroke separately from hemorrhagic stroke to avoid masking the true effect of POI on stroke. The relationship between POI, ENM and higher risk of all-cancer mortality was also not found in our analysis. Studies indicate that early menopause increased risk of mortality from digestive tract cancer[13,35], but decreased risk of mortality from breast, uterine and ovarian cancer[11,35]. Estrogen may play an opposing role in the influence on digestive tract cancer and breast, uterine, and ovarian cancers. It has a combined effect on all-cancer mortality in postmenopausal women.

The factors of hormone therapy and smoking are the two major confounders in the association of ANM and risk of death. The timing hypothesis states that there are significant trends in an effect of hormone therapy on IHD in women according to age and time since menopause. Hormone therapy reduces both IHD events and total mortality in primary prevention when it is initiated in women who are within 10 years of menopause and under the age of 60 years[31]. The association between ENM and mortality outcomes may be underestimated if the hormone therapy factor is not controlled for. It may partly explain why no relationship of POI and risk from all-cause death is found in the group in which subgroup analysis did not control for the hormone therapy factor. Smoking as another important confounder is related most consistently to early menopause[36], and increases death rates from several major causes such as cardiovascular disease and cancer.

The association between early menopause and mortality outcomes may be overestimated if the smoking factor is not controlled for. But no relationship of POI with risk from all-cause death in this review was found in the group in which the smoking factor was not controlled for, only based on one study.

Moreover, subgroup analyses in our review indicated that there was no significant relationship between POI and all-cause mortality in Norwegians, but a significant relationship between POI and all-cause mortality was found in Japanese and Americans. This suggested that racial difference was likely to be a potential moderating factor. In further studies, this underlying difference should be taken into account. Publication bias in reports of all-cause mortality might be partly explained by this difference.

This meta-analysis was based on several prospective cohort studies from various populations. The combined sample size was large and the follow-up period was long in most studies. The estimates from the adjusted models for each study were used in our analyses to reduce the potential of confounders. Our results may have potential public health implications. The contribution of reproductive history, in particular spontaneous menopause before the age of 40 years, for disease risk prediction should be considered in future studies.

There were, however, several limitations in our meta-analysis. Due to the self-reported age of menopause, an exposure misclassification was likely to bias the effect estimates toward the null and to attenuate the possible relationship[37]. Although the potential confounders were considered in most included studies, the residual confounding from unmeasured socioeconomic or other potential factors in prospective cohort studies still cannot be excluded. Conversely, the most comprehensively adjusted models that controlled for adult body mass index and chronic conditions might be susceptible to over-adjustment; these factors could lie on the causal pathway between early menopause and mortality outcomes.

The cut-off point for POI and ENM and the age stages of reference for normal ANM were in disparity among some studies and might have led to underestimate or overestimate outcomes; meanwhile, the dose–response analysis could not be conducted due to the limitation of primary data in studies. Due to the changes in International Classification of Diseases codes from the ninth to the tenth revision, the discrepancies in the definitions of CVD deaths were a potential source of heterogeneity and misclassification. In this review, some outcomes mainly represented the United States, thus limiting the generalizability of the results. Finally, the potential for publication bias remains since the search was limited to all English-language published studies.

Conclusion

The current meta-analysis provides evidence for a positive association between POI and IHD and all-cause mortality. No significant association was detected from comparisons of stroke and all-cancer mortality in POI women and normal ANM women. ENM was only associated with a slightly higher risk of death from IHD. More prospective studies are still needed to clarify the association between ANM and mortality and to confirm the potential dose–response relationship.

Conflict of interest The authors report no conflict of interest. The authors alone are responsible for the content and writing of this paper.

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